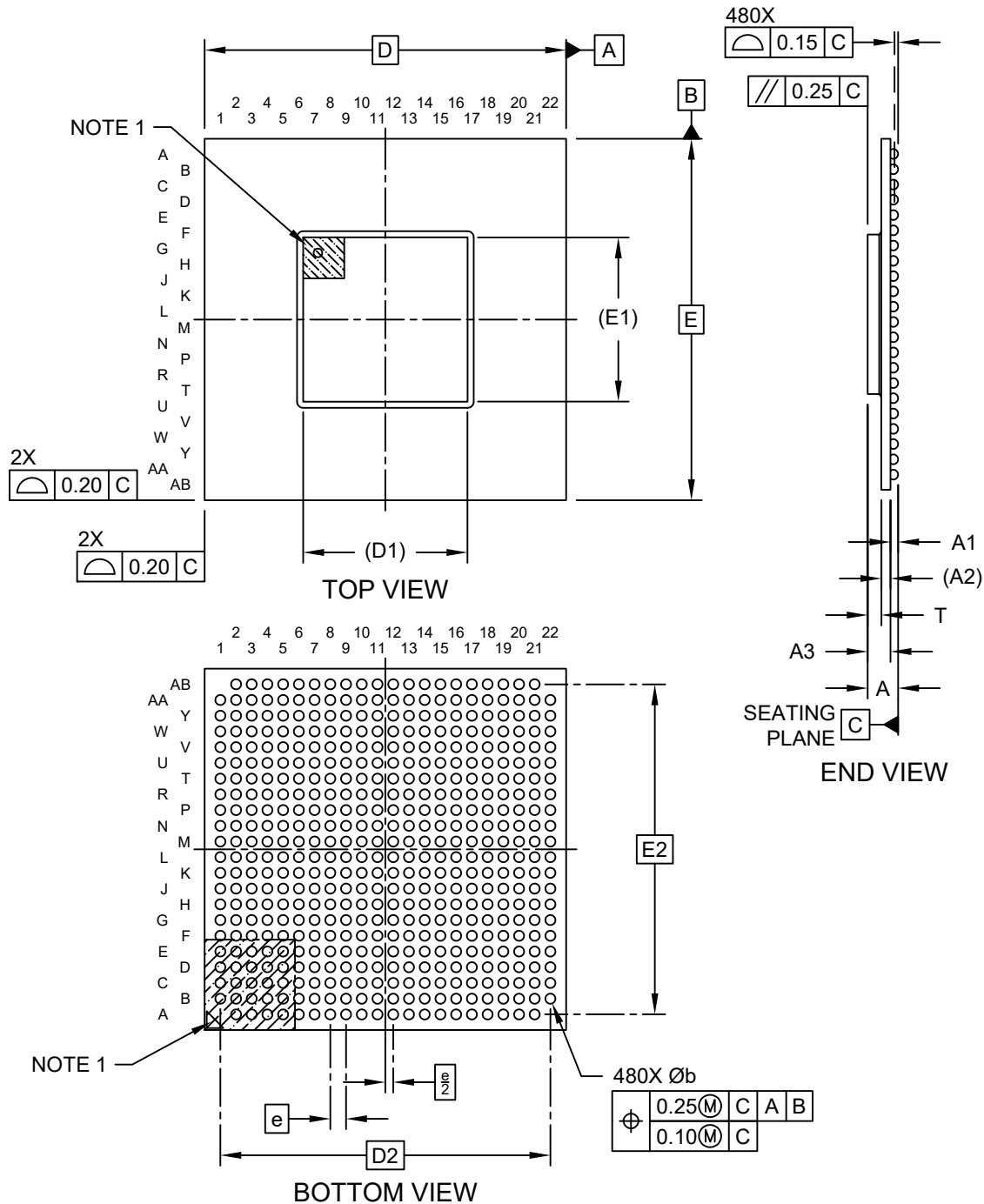


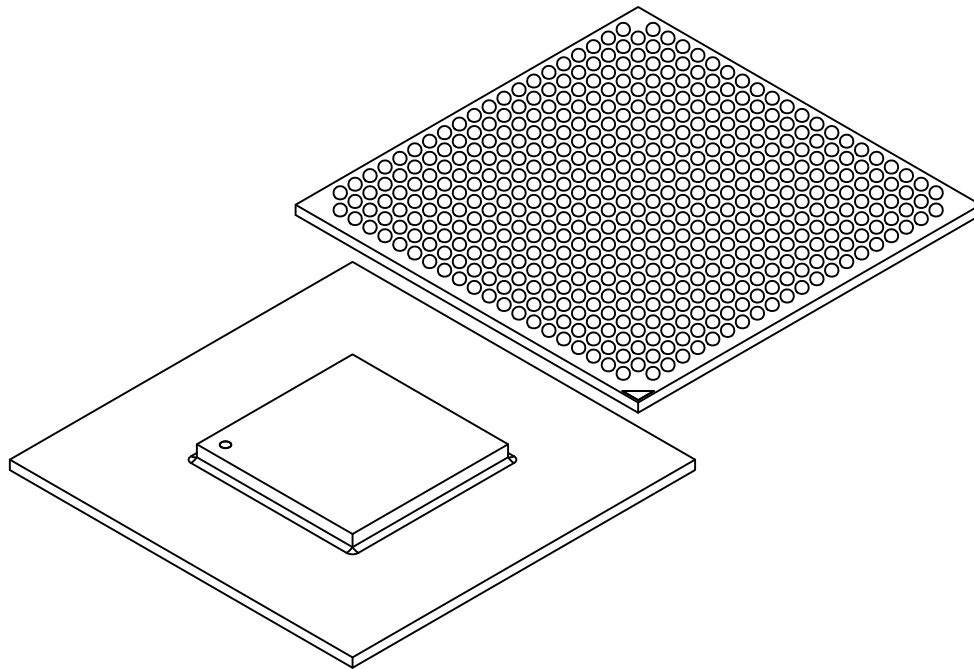
480-Ball [Standard] Ball Grid Array (9KC) - 23x23x2.24 mm Body [BGA] Flip Chip; Microsemi Legacy Package

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



480-Ball [Standard] Ball Grid Array (9KC) - 23x23x2.24 mm Body [BGA] Flip Chip; Microsemi Legacy Package

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



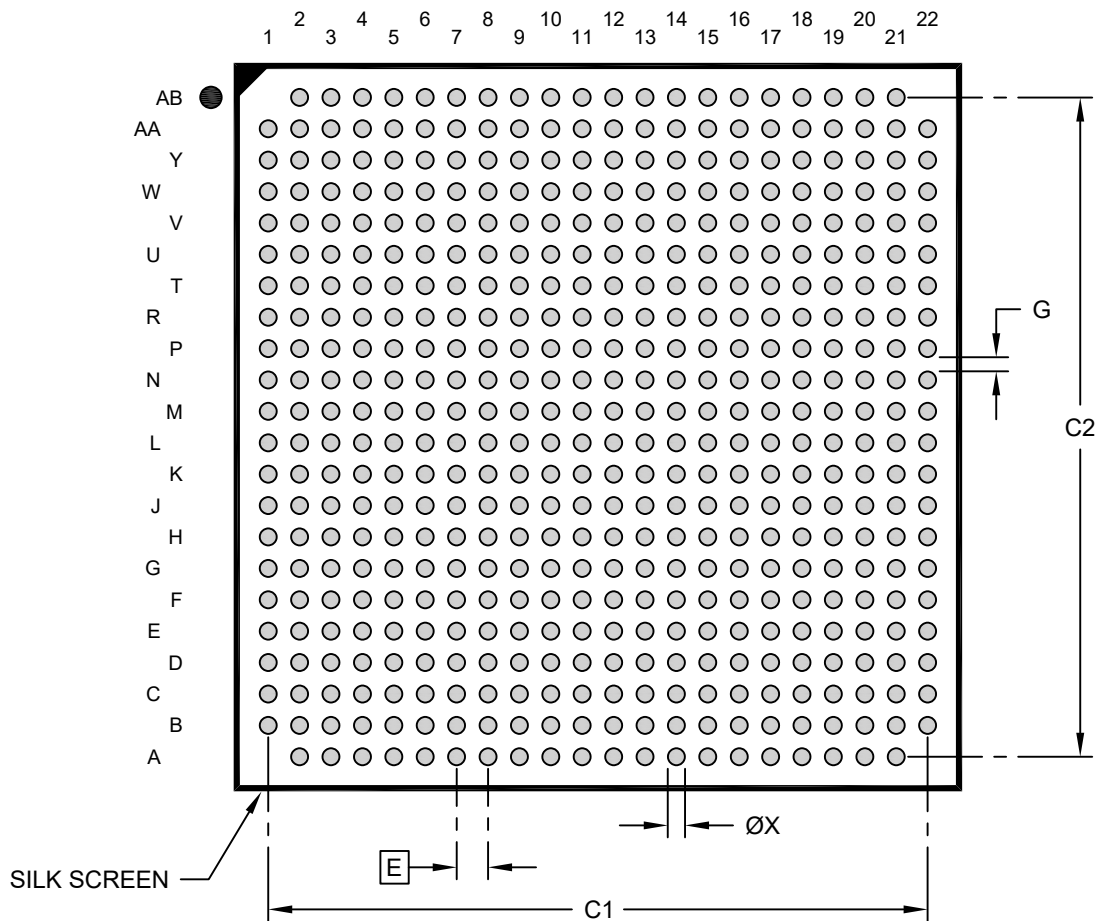
		Units	MILLIMETERS		
Dimension Limits			MIN	NOM	MAX
Number of Terminals	N		480		
Pitch	e		1.00 BSC		
Overall Height	A		1.68	1.96	2.24
Ball Height	A1		0.40	0.50	0.60
Substrate Thickness	A2		0.60		
Package Thickness	A3		1.28	1.46	1.64
Die Thickness	T		0.875	0.90	0.925
Overall Length	D		23.00 BSC		
Overall Length	D1		10.45 REF		
Ball Array Length	D2		21.00 BSC		
Overall Width	E		23.00 BSC		
Overall Width	E1		10.45 REF		
Ball Array Width	E2		21.00 BSC		
Ball Diameter	b		0.50	0.64	0.70

Notes:

- Terminal A1 visual index feature may vary, but must be located within the hatched area.
- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

480-Ball [Standard] Ball Grid Array (9KC) - 23x23x2.24 mm Body [BGA] Flip Chip; Microsemi Legacy Package

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	1.00 BSC		
Contact Pad Spacing	C1		21.00	
Contact Pad Spacing	C2		21.00	
Contact Pad Diameter	X			0.55
Contact Pad to Contact Pad (Xnn)	G2	0.45		

Notes:

1. Dimensioning and tolerancing per ASME Y14.5M

BSC: Basic Dimension. Theoretically exact value shown without tolerances.

Microchip Technology Drawing C04-27139 Rev A